Glossary — Days 1 to 4

Saturday 24 Oct 2026 · Every abbreviation and key term used so far, with a plain-English meaning and where you met it

How to use this
Skim a section before revising the matching day's note. If a term in a question feels fuzzy, find it here first. From Day 5 onwards every term is explained in the lesson itself, and each day's note ends with its own glossary.

1. AI and model basics

Term	Full form	Meaning	Example / where you met it
AI	Artificial Intelligence	Software that performs tasks needing human-like judgement.	The umbrella term for everything in this exam.
GenAI	Generative AI	AI that creates new content (text, images, code) instead of only classifying.	A chatbot writing a refund email.
LLM	Large Language Model	A model trained on huge amounts of text that predicts the next words.	Claude, Amazon Nova, Llama.
FM	Foundation Model	A large, general-purpose pre-trained model you build applications on.	Amazon Bedrock is a catalogue of FMs.
Token	—	A chunk of text (roughly ¾ of a word) that models read and write; billing and limits count tokens.	'Hello world' is about 2–3 tokens.
Context window	—	The maximum number of tokens a model can consider in one request.	Why you can't paste 10,000 documents into one prompt.
Hallucination	—	A confident answer that isn't supported by facts or sources.	Inventing a refund amount not in the policy.
Embedding	—	A list of numbers representing the meaning of text; similar meanings give nearby vectors.	'car broke down' sits near 'vehicle failure'.
Vector	—	The list of numbers produced by an embedding model.	Titan V2 produces 256/512/1024 numbers.
Fine-tuning	—	Further training a model on your examples to change its behaviour or style.	Teaching a model your company's writing tone.
Inference	—	Running a trained model to get an output (as opposed to training it).	Every InvokeModel / Converse call.
Temperature	—	Setting controlling randomness of output; low = more deterministic.	0 for extraction tasks, higher for brainstorming.
LLM-as-a-judge	—	Using a model to score another model's outputs against criteria.	Bedrock Evaluations scoring context relevance.

2. RAG and retrieval (Day 1)

Term	Full form	Meaning	Example / where you met it
RAG	Retrieval-Augmented Generation	Search your own data first, then give the relevant passages to the model so it answers from them.	A support bot answering from company policy PDFs.
KB	Knowledge Base (Amazon Bedrock Knowledge Bases)	AWS's managed RAG service: ingests, chunks, embeds, stores and retrieves your documents.	Day 1 lab: KB over 5 PDFs in S3.
Data source	—	The location a Knowledge Base ingests from (e.g. an S3 bucket); chunking is set here.	Not to be confused with a data store.
Sync / ingestion job	—	The process that reads a data source and updates the KB index.	StartIngestionJob, scheduled hourly.
Chunking	—	Splitting documents into smaller passages before embedding.	Fixed-size, hierarchical, semantic, none, custom (Lambda).
Hierarchical chunking	—	Small 'child' chunks are searched; their larger 'parent' chunk is returned.	Precise match plus full surrounding context.
Semantic search	—	Search by meaning using vectors.	Finds synonyms and paraphrases.
BM25	Best Matching 25	A classic keyword-ranking algorithm (exact word matching).	Finds 'XJ-4471-B' exactly.
Hybrid search	—	Combines semantic and keyword (BM25) search.	Fix for missed product codes.
RRF	Reciprocal Rank Fusion	A method of merging two ranked result lists into one.	Azure AI Search hybrid queries.
Reranker / rerank model	—	A second model that re-scores retrieved chunks so the best is first.	Amazon Rerank, Cohere Rerank; Azure semantic ranker.
HNSW	Hierarchical Navigable Small World	A fast algorithm for finding nearest vectors.	Used inside vector databases.
kNN	k-Nearest Neighbours	Finding the k vectors closest to a query vector.	Vector search.
Metadata filtering	—	Restricting retrieval to chunks whose attributes match (tenant, department).	Multi-tenant isolation.
GraphRAG	Graph-based RAG	RAG that uses relationships between entities stored in a graph.	Neptune Analytics as the vector store.
Retrieve vs RetrieveAndGenerate	—	Retrieve returns chunks only; RetrieveAndGenerate returns a full answer with citations.	Custom logic in between → Retrieve.
OCR	Optical Character Recognition	Reading text from images or scanned pages.	Amazon Textract.
BDA	Amazon Bedrock Data Automation	Managed extraction of structured data from documents, images, audio, video.	Parsing table-heavy PDFs for a KB.
Grounding / groundedness	—	Whether an answer is supported by the retrieved sources.	Guardrails contextual grounding check.
Context precision / recall	—	Precision: are retrieved chunks relevant? Recall: was everything needed retrieved?	RAG evaluation metrics.

3. Agents and tools (Days 2–3)

Term	Full form	Meaning	Example / where you met it
Agent	—	A model plus instructions, tools and memory running in a loop, deciding at runtime which tool to call.	OrderHelper checking then cancelling an order.
ReAct	Reason + Act	The agent loop pattern: think → call a tool → observe → repeat.	Default Bedrock Agents orchestration.
Tool / function calling	—	The model requests a named tool with arguments; your code runs it.	stopReason = tool_use.
toolUse / toolResult	—	Converse API blocks: the model's tool request, and your reply with the output (matched by toolUseId).	Results go in a role 'user' message.
Action group	—	Bedrock Agents Classic's way of defining tools (OpenAPI schema or function details).	get_order_status, cancel_order.
RoC	Return of control	Bedrock Agents hands the tool call back to your application to execute.	Banking app using the user's authenticated session.
DRAFT / version / alias	—	DRAFT is the editable agent; Prepare → version (snapshot); an alias points to a version.	Rollback = repoint the alias.
TSTALIASID	Test alias ID	Built-in alias that points to the DRAFT version, for testing.	Test after PrepareAgent.
HITL	Human-in-the-loop	A person approves or reviews a step before it proceeds.	requireConfirmation; Step Functions approval.
Agents Classic	Amazon Bedrock Agents (Classic)	The original managed agents service, now in maintenance mode for new accounts.	Why your Day 2 console lab was blocked.
AgentCore	Amazon Bedrock AgentCore	AWS services to host, connect, secure and observe agents built with any framework.	Runtime, Gateway, Memory, Identity, Observability.
Strands Agents	—	AWS's open-source Python SDK for building agents.	@tool decorator.
Agent Squad	AWS Agent Squad (formerly Multi-Agent Orchestrator)	Open-source framework that routes each request to the best specialist agent.	Billing / tech / sales routing.
MCP	Model Context Protocol	Open standard for connecting AI apps to tools and data ('USB-C for AI tools').	Tools exposed once, usable by any framework.
A2A	Agent-to-Agent protocol	Open standard for agents to communicate with other agents.	Mentioned for Azure/AI-500.
Supervisor / router	—	Supervisor: delegates to several agents and combines. Router: sends to exactly one.	95% single-specialist traffic → router.
Swarm	—	Peer agents hand work to each other without a central coordinator.	Exploratory tasks.
microVM	Micro virtual machine	A tiny, fast-starting isolated VM.	AgentCore Runtime session isolation.
OpenTelemetry (OTel)	—	Open standard format for traces, metrics and logs.	AgentCore Observability.
Code Interpreter	—	A sandbox where an agent runs code it writes.	pandas analysis of a CSV.

4. AWS services named so far

Term	Full form	Meaning	Example / where you met it
AWS	Amazon Web Services	Amazon's cloud platform.	—
S3	Amazon Simple Storage Service	Object storage for files of any kind.	KB source documents.
Lambda	AWS Lambda	Runs your code on demand without servers (max 15 minutes per run).	Action group executor, Gateway target.
Step Functions	AWS Step Functions	Visual workflows that chain steps with retries and history.	Mortgage approval workflow.
Standard vs Express	Step Functions workflow types	Standard: up to 1 year, full history. Express: up to 5 minutes, high volume.	Multi-day approval → Standard.
waitForTaskToken	—	Step Functions pattern that pauses until someone returns a task token.	Human approval via SQS + SendTaskSuccess.
SQS	Amazon Simple Queue Service	Message queue between components.	Delivering the approval token.
SNS	Amazon Simple Notification Service	Pub/sub notifications (email, SMS, HTTP).	Alerting.
EventBridge	Amazon EventBridge	Event bus and scheduler that triggers targets on events or schedules.	Hourly StartIngestionJob.
API Gateway	Amazon API Gateway	Managed front door for HTTP/REST APIs.	Exposing a GenAI app.
ECS / Fargate	Elastic Container Service / serverless containers	Run containers; Fargate removes server management.	Distractor vs AgentCore Runtime.
ALB	Application Load Balancer	Distributes HTTP traffic across targets.	Distractor in Q11.
OpenSearch Serverless	Amazon OpenSearch Serverless	Managed search/vector engine without clusters.	Default KB vector store.
OCU	OpenSearch Compute Unit	Billing unit for OpenSearch Serverless capacity, charged hourly.	Why you delete it after labs.
Aurora / pgvector	Amazon Aurora PostgreSQL + pgvector extension	Relational database with vector search support.	Vector store when you need SQL joins.
Neptune Analytics	Amazon Neptune Analytics	Graph analytics database.	GraphRAG.
DynamoDB	Amazon DynamoDB	Key-value / document NoSQL database. **Not** a KB vector store.	Practice Set Q2 trap.
Textract	Amazon Textract	Extracts text, forms and tables from scans (OCR).	Scanned claim forms.
Comprehend	Amazon Comprehend	Natural-language service; detects and redacts PII.	Redact before ingestion.
Macie	Amazon Macie	Scans S3 to discover sensitive data such as PII.	Find card numbers in buckets.
Glue / Data Catalog	AWS Glue	Data integration (ETL) and a catalogue of datasets/tables.	Tables governed by Lake Formation.
Lake Formation	AWS Lake Formation	Fine-grained table/column/row permissions for data lakes.	Hide card_number column.

Term	Full form	Meaning	Example / where you met it
CloudWatch	Amazon CloudWatch	Metrics, logs, dashboards and alarms.	GenAI Observability dashboard.
CloudTrail	AWS CloudTrail	Records who called which AWS API, and when.	Audit of Bedrock calls.
X-Ray / Transaction Search	AWS X-Ray, CloudWatch Transaction Search	Distributed tracing; Transaction Search must be enabled for AgentCore traces.	Day 3 console lab Part D.
Secrets Manager	AWS Secrets Manager	Stores and rotates secrets (passwords, API keys).	AgentCore Identity API-key provider.
Cognito	Amazon Cognito	User sign-up/sign-in; issues JWTs.	Gateway inbound auth quick-create.
CloudShell	AWS CloudShell	Browser terminal with the AWS CLI pre-configured.	send-task-success.
Service Quotas	—	Console for viewing and raising account limits.	Bedrock TPM quota of 0.

5. Security and networking (Day 4)

Term	Full form	Meaning	Example / where you met it
IAM	AWS Identity and Access Management	Who (users, roles) can do what (actions) on which resources.	Allow InvokeModel on one model only.
IAM role	—	An identity with permissions that services or apps assume, giving short-term credentials.	KB service role.
Identity-based vs resource-based policy	—	Identity policy is attached to the caller; resource policy is attached to the resource saying who may use it.	Lambda resource policy allowing bedrock.amazonaws.com.
ARN	Amazon Resource Name	The unique ID string of an AWS resource.	arn:aws:bedrock:us-east-1::foundation-model/amazon.nova-lite-v1:0
Implicit vs explicit deny	—	Implicit: nothing allowed it. Explicit: a Deny statement — always wins.	Policy Simulator results.
SCP	Service Control Policy	Organisation-wide guardrail capping what any member account (even admins) can do.	Block one model across the company.
AWS Organizations	—	Groups many AWS accounts under central management; where SCPs live.	—
KMS	AWS Key Management Service	Creates and controls encryption keys; every use is auditable.	Encrypt KB data with your own key.
CMK / customer managed key	Customer managed KMS key	A KMS key you create and control (rotation, policy, revocation).	'Control and audit the keys'.
Key policy	—	The resource policy on a KMS key saying who may use it.	Must allow KB role kms:Decrypt.
SSE	Server-Side Encryption	The service encrypts data when storing it.	S3 default encryption.

Term	Full form	Meaning	Example / where you met it
SSE-S3 / SSE-KMS	—	SSE-S3: keys managed by S3. SSE-KMS: encryption using a KMS key you can control.	Switching to SSE-KMS broke the KB sync.
VPC	Virtual Private Cloud	Your private, isolated network inside AWS.	Private subnets.
Subnet (private)	—	A slice of a VPC with no direct internet route.	Healthcare app in Q1.
NAT gateway	Network Address Translation gateway	Lets private resources reach the internet outbound.	Not needed when you use VPC endpoints.
VPC endpoint / VPCE	VPC Endpoint	A private connection from your VPC to an AWS service, no internet.	bedrock-runtime, bedrock-agent-runtime endpoints.
PrivateLink	AWS PrivateLink	The technology behind interface VPC endpoints.	—
aws:SourceVpce	—	IAM condition key: which VPC endpoint the request came through.	Deny if not our endpoint.
Endpoint policy	—	A policy on a VPC endpoint limiting what can be called through it.	Only Nova models.
DNS / private DNS	Domain Name System	Translates service names into addresses; private DNS makes the normal service name resolve to your endpoint.	Enable when creating the endpoint.
PII	Personally Identifiable Information	Data that identifies a person.	Names, emails, card numbers.
Mask vs block	Guardrails PII actions	Mask replaces the PII with a placeholder; block rejects the whole input/output.	Email → mask, card → block.
Regex	Regular expression	A text pattern for matching strings.	ORD-\d{6} for order IDs.
WORM / Object Lock	Write Once, Read Many	Data can't be changed or deleted for a set period.	Compliance retention.
Lifecycle rule	S3 Lifecycle	Automatically moves or deletes objects after N days.	Expire logs after 30 days.
Data residency	—	Requirement that data stays in a specific country/region.	Use a geographic cross-region inference profile.
CRIS	Cross-Region Inference	Bedrock routes requests to other regions for capacity.	us.* inference profile IDs.
Invocation logging	Model invocation logging	Optional Bedrock feature recording full prompts and responses to S3/CloudWatch.	Encrypt with KMS, expire with lifecycle.

6. Authentication, APIs and developer terms

Term	Full form	Meaning	Example / where you met it
API	Application Programming Interface	A defined way for programs to call each other.	Converse API.
SDK	Software Development Kit	Libraries for calling a service from code.	boto3 (Python AWS SDK).

Term	Full form	Meaning	Example / where you met it
boto3	—	The AWS SDK for Python.	brt.converse(...).
CLI	Command Line Interface	Terminal tool for calling a service.	aws stepfunctions send-task-success.
CDK	AWS Cloud Development Kit	Define infrastructure in code (TypeScript, Python...).	—
REST / HTTP	Representational State Transfer	Common style of web APIs over HTTP.	300 internal REST APIs.
OpenAPI	OpenAPI Specification	A standard file describing a REST API's endpoints and parameters.	Action group schema, Gateway target.
JSON	JavaScript Object Notation	Text format for structured data.	Tool input schemas.
JSONPath / JSONata	—	Two expression languages Step Functions uses to read and transform data.	$$.Task.Token vs $states.context.Task.Token.
OAuth	Open Authorization	Standard for letting an app act on a user's behalf without their password.	AgentCore Identity outbound to GitHub.
JWT	JSON Web Token	A signed token proving who the caller is.	Cognito-issued token for the Gateway.
Inbound vs outbound auth	—	Inbound: who may call the agent/gateway. Outbound: how the agent calls other services.	Cognito JWT in; IAM role/OAuth out.
PAT	Personal Access Token	A long-lived token tied to one user (e.g. GitHub).	Q12 distractor.
TPM / RPM	Tokens / Requests Per Minute	Rate limits on model usage.	Your quota was 0 TPM.
Throttling	ThrottlingException	Error when you exceed a rate limit.	Fix: backoff, quota increase.
Exponential backoff	—	Retry with growing waits (1s, 2s, 4s...).	Handling throttling.
ETL	Extract, Transform, Load	Moving and reshaping data between systems.	Ingestion pipelines.
Pass / Choice / Task state	Step Functions states	Pass: placeholder; Choice: branch; Task: do work.	Day 3 console lab Part E.

7. Exam and Azure terms

Term	Full form	Meaning	Example / where you met it
AIP-C01	—	Exam code for AWS Certified Generative AI Developer – Professional.	Mon 9 Nov.
AI-103	—	Microsoft exam: Developing AI Apps and Agents on Azure.	Sat 28 Nov.
AI-500	—	Microsoft exam: Designing and Implementing Multi-Agent AI Solutions (Expert).	December.
Foundry	Microsoft (Azure AI) Foundry	Azure's platform for models and agents.	Azure equivalent of Bedrock.

Term	Full form	Meaning	Example / where you met it
Agent Service	Foundry Agent Service	Azure's managed agent runtime (threads, messages, runs).	requires_action = return of control.
AI Search	Azure AI Search	Azure's search/vector service.	Azure equivalent of KB vector store.
Content Safety	Azure AI Content Safety	Azure content filtering incl. groundedness detection.	Azure equivalent of Guardrails.
Entra ID	Microsoft Entra ID	Azure's identity service (formerly Azure AD).	Azure equivalent of IAM users.
ESL +30	English as a Second Language +30 minutes	AWS exam accommodation (not requested).	—